

**SEMESTER 2
2021-2022**

**CS426
Computer Graphics**

Dr. C. Gurrin, Dr. J. Timoney, Dr. C. Markham

Time allowed: 2 hours

Answer all **four** questions

Your mark will be based on your **four** answers

All questions carry equal marks

Instructions

	Yes	No	N/A
Formulae and Tables book allowed (<i>i.e. available on request</i>)		X	
Formulae and Tables book required (<i>i.e. distributed prior to exam commencing</i>)		X	
Statistics Tables and Formulae allowed (<i>i.e. available on request</i>)		X	
Statistics Tables and Formulae required (<i>i.e. distributed prior to exam commencing</i>)		X	
Dictionary allowed (<i>supplied by the student</i>)		X	
Non-programmable calculator allowed		X	
Students required to write in and return the exam question paper		X	

[25 marks]

- 1 (a) Calculate numerical values for a and b of the 2D homogeneous co-ordinate represented by v . [4 marks]

$$v = \begin{bmatrix} 6 \\ 9 \\ 3 \end{bmatrix} \quad v = ai + bj$$

Calculate the angle the vector makes with respect to the i -axis.

- (b) Consider the frame shown below and answer the questions. [5 marks]

$$P = \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & b & 0 \\ \frac{1}{2} & a & \frac{1}{\sqrt{2}} & 4 \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} & 3 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

Calculate the missing values, a and b .

Calculate the distance the frame is from the origin.

- (c) Match the following matrices with the best description from the list below. [8 marks]

A	B	C	D
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$
E	F	G	H
$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Q	Rotation 90° about Z axis	V	Rescale all axes
R	Translate 1 in Z direction	W	Projective transformation
S	Rotation 90° about X axis	X	Orthographic projection
T	Identity matrix – no change	Y	Rescaling of the Y axis only
U	Reflection in the YZ plane	Z	No descriptions apply

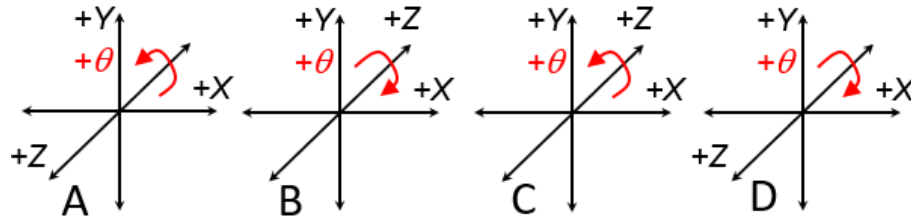
- (d) Consider the phrase “the dot product is fundamental to computer graphics” and answer the following questions. [8 marks]

(i) Provide a mathematical description of the 2D dot product. This should reference measuring the angle between two vectors.

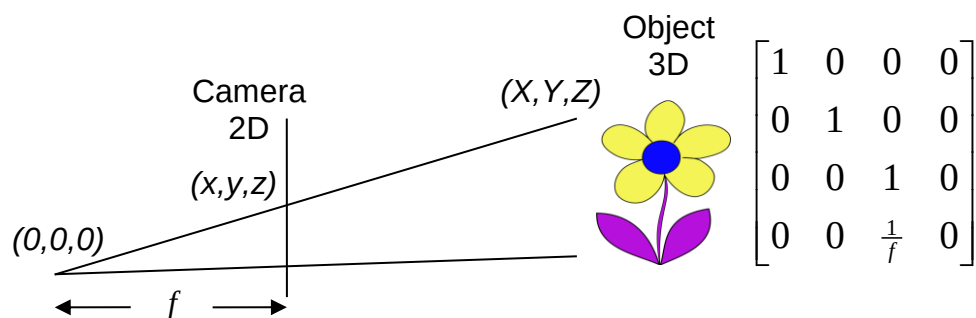
(ii) List three areas of graphics where the dot product has been central to the discussion. Explain your choice.

[25 marks]

- 2 (a) Identify the left-handed and right-handed coordinate systems in the set of options shown below (if they exist). [4 marks]



- (b) A camera can be modeled using a projective transformation that maps points in 3D (X,Y,Z) to the screen which is 2D. Derive expressions for this transformation and show that in Homogeneous form the following matrix implements this transformation. Comment on how this projective transformation is different to that used by OpenGL. [8 marks]



- (c) (i) Evaluate the following quaternion product. [6 marks]

$$(1 + 1i + 0j + 2k) \cdot (1 + 0i + 2j + 0k)$$

- (ii) Calculate the values for a quaternion representing a 45-degree rotation about the Y-axis. Express your answer in the form

$$r = A + Bi + Cj + Dk$$

- (iii) Describe the method you would use to rotate a point P about the quaternion r.

- (d) Distinguish between the terms ray tracing and ray casting. Describe each of the rays that are used as part of a ray tracing algorithm. [7 marks]

[25 marks]

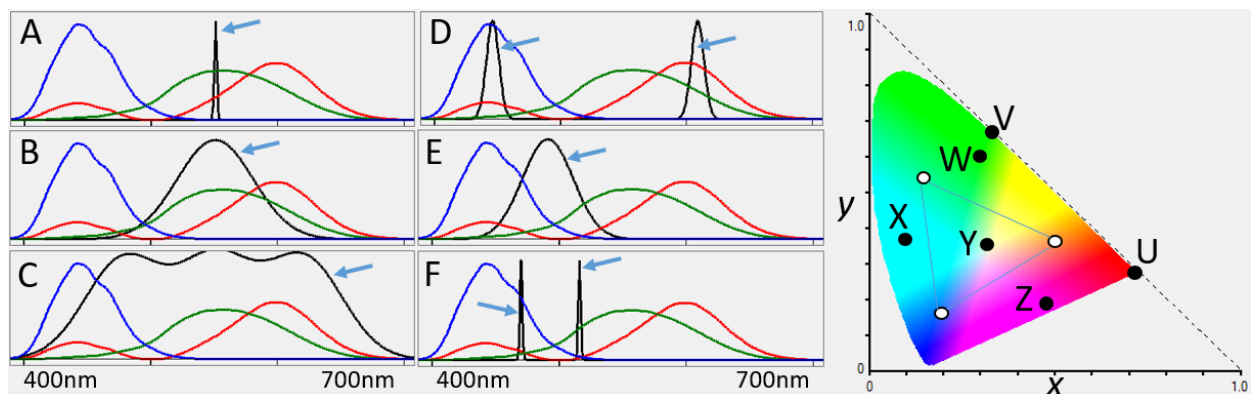
- 3 (a) Consider the object wavefront file shown below and answer the questions that follow. [7 marks]

exam.obj	exam.mtl	bird.bmp
mtllib exam.mtl	newmtl m1	
o Object	Ka 0.0 0.0 1.0	
v 0.0 0.0 -1.0	Kd 0.0 0.0 1.0	
v 0.866 0.0 0.5	Ks 0.0 0.0 1.0	
v -0.866 0.0 0.5		
v 0.0 1.0 0.0	newmtl m2	
vt 0.0 0.0	Ka 1.0 0.0 0.0	
vt 0.5 1.0	Kd 1.0 0.0 0.0	
vt 1.0 0.0	Ks 1.0 0.0 0.0	
usemtl m1		
f 1 4 2	newmtl m3	
f 2 4 3	Ka 1.0 1.0 1.0	
usemtl m2	Kd 1.0 1.0 1.0	
f 3 4 1	Ks 1.0 1.0 1.0	
usemtl m3	map_Kd bird.bmp	
f 1/1 2/2 3/3		

- Which platonic solid do the files represent?
 - How many faces are red? Explain your reasoning.
 - Use a drawing to show how the bird is rendered on one of the faces of the object. Use this to explain UV mapping.
- (b) Use a mathematical argument to describe how to convert a set of RGB values describing a red object into the corresponding CMY values. Explain your reasoning. [4 marks]

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- (c) Consider the follow set of images showing the relationship between a spectral response and the corresponding point on the chromaticity diagram. Answer the questions that follow. [7 marks]



- (i) Match the spectral response with the corresponding colour identified on the chromaticity diagram (e.g. AZ, BY...)
 - (ii) What is the name given to two different spectra that appear the same colour?
 - (iii) Name the triangular region represented by the three white points? What relevance does this region have to the performance of a monitor?
- (d) Consider the following transformations from the OpenGL graphics pipeline and answer the questions that follow. [7 marks]

A: Projection Transform

B: Viewport Transform

C: World Transform

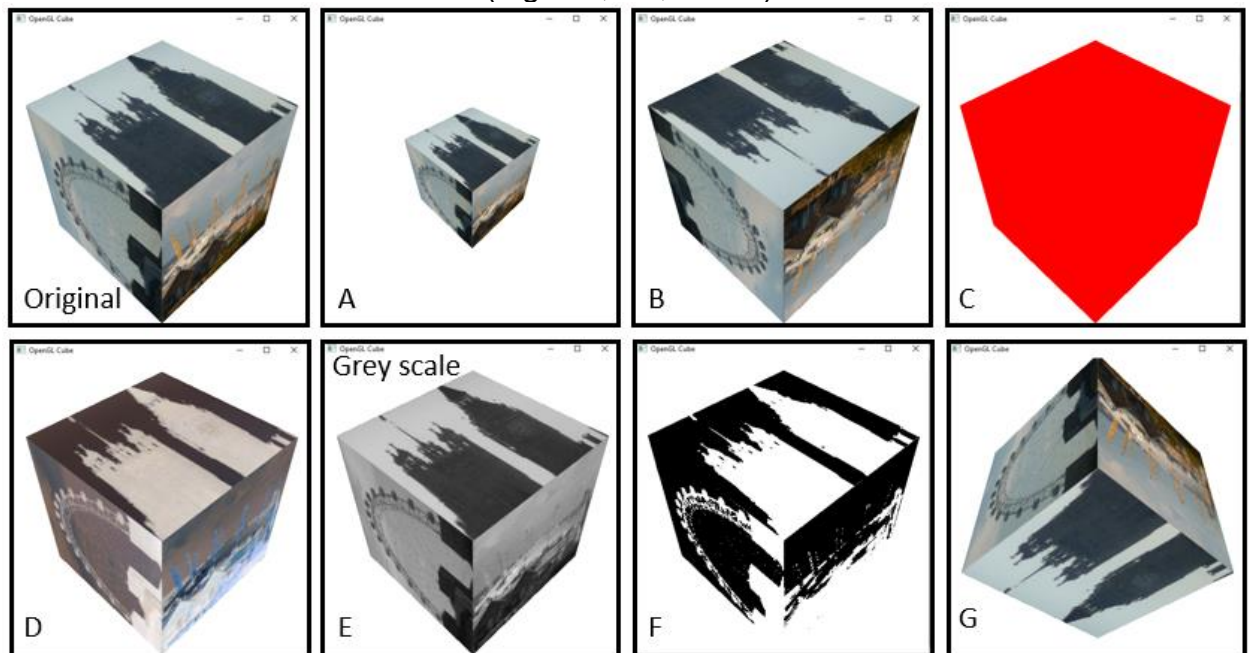
D: View Transform

E: Perspective Divide

- (i) Briefly describe each transformation (one sentence each).
- (ii) Identify the order in which the computer will process these transformations (e.g. EDCBA).

[25 marks]

- 4 (a) Consider the images shown below and match them with shader code that created them (e.g. AS, BZ, CV....). [7 marks]



```
#version 330 core
#extension GL_ARB_separate_shader_objects:enable
layout(location = 0) in vec3 vertex_modelspace;
layout(location = 1) in vec2 vertexUV;
out vec2 UV;
uniform mat4 MVP;

void main(){
    mat4 M;
    //M[0]=vec4(1,0,0,0);
    //M[1]=vec4(0,-1,0,0);
    //M[2]=vec4(0,0,1,0);
    //M[3]=vec4(0,0,0,1);
    //gl_Position=M*MVP*vec4(vertex_modelspace,1);

    //M[0]=vec4(0.5,0,0,0);
    //M[1]=vec4(0,0.5,0,0);
    //M[2]=vec4(0,0,0.5,0);
    //M[3]=vec4(0,0,0,1);
    //gl_Position=M*MVP*vec4(vertex_modelspace,1);

    //gl_Position=MVP*vec4(vertex_modelspace,1);

    UV = vertexUV;
    //UV = vec2(vertexUV.s, 1.0 - vertexUV.t);
}
```

```
#version 330 core
in vec2 UV;
out vec3 color;
uniform sampler2D myTextureSampler;
void main(){
    color = texture( myTextureSampler, UV ).rgb;
    //color=vec3(1-color.r,1-color.g,1-color.b); }-W
    //float i=color.r*0.3+color.g*0.6+color.b*0.1; }-X
    //color=vec3(i,i,i);
    //float m = (color.r+color.g+color.b)/3.0; }-Y
    //if (m>0.5) m=0; else m=1;
    //color=vec3(m,m,m);
    //color = vec3(1,0,0); }-Z
}
```

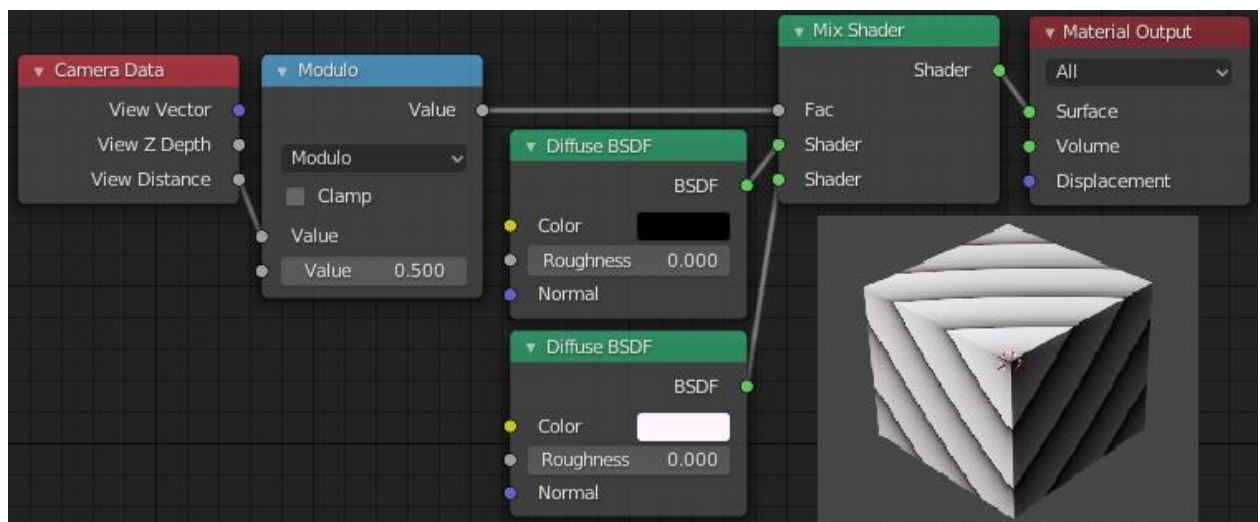
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- (b) Consider the following equation and answer the questions that follow. [6 marks]

$$I(x, y) = \underbrace{k_a I_A(x, y)}_A + I_l(x, y) \left(\underbrace{k_d (\vec{L} \cdot \vec{N})}_B + \underbrace{k_s (\vec{R} \cdot \vec{V})^n}_C \right)$$

- Name the illumination model?
- Name and describe each of types of illumination represented by A, B and C.
- Explain the vectors L, N, R and V. Include a diagram.

- (c) Consider the Blender shader shown below and answer the questions that follow. [6 marks]



- Describe the function blocks and the data communicated between in the shader code above. [6 marks]
- Explain how the shader creates the cube texture shown.
- Explain the acronyms BSDF, BRDF and BTDF.

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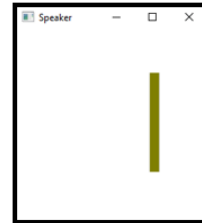
- 4 (d) The following differential equation was used to describe the motion of a loudspeaker cone subject to an input signal. [6 marks]

$$0 = i + K \frac{d^2x}{dt^2} + D \frac{dx}{dt} + Hx$$

Using Euler's method a computer simulation was created to evaluate the above expression. Answer the questions that follow.

```
#include <freeglut.h>
#include <stdarg.h>
#include <stdio.h>
#include <stdlib.h>
float i = -2.0, x = 0.0, v = 0.0;
float K = 1.0, D = 0.5, H = 5.0, dt = 0.1;
static void display(void)
{
    glClear(GL_COLOR_BUFFER_BIT);
    glClearColor(1, 1, 1, 1);
    glColor3f(0.5, 0.5, 0.0);
    glRectf(x-0.05, -0.5, x+0.05, 0.5);
    glutSwapBuffers();
}
static void idle()
{
    x = ... Fill in the missing lines
    v = ...
    glutPostRedisplay();
}

int main(int argc, char* argv[])
{
    glutInit(&argc, argv);
    glutInitWindowSize(256, 256);
    glutInitWindowPosition(100, 100);
    glutInitDisplayMode(GLUT_RGB|GLUT_DOUBLE|
        GLUT_DEPTH|GLUT_MULTISAMPLE);
    glutCreateWindow("Speaker");
    glMatrixMode(GL_PROJECTION);
    gluOrtho2D(-1.0, 1.0, -1.0, 1.0);
    glutDisplayFunc(display);
    glutIdleFunc(idle);
    glutMainLoop();
    return 0;
}
```



- (i) Explain the physical significance of x , K , D and H .
- (ii) Complete the two missing lines of code.
- (iii) Explain the term call-back function and identify them in the code above.